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Homework 4

1. The Java type you chose to represent the adjacency list inside your graph class.

I used a hash map where each key is a vertex and the value is a list of its edges.

2. How you assigned edge weights when converting the image to a graph.

I simply assigned the weight by the numbers the Bitmap Processor class method getRGBMatrix gave me and multiplied those numbers by -1 to make them positive.

3. How you solved the proposed problem using a constant number of invocations of Dijkstra's algorithm.

I solved this by having a top node connected to all the vertices at the top of the graph and having a left node connected to all the vertices on the left of the graph. This made it so I only have to call Dijkstra's algorithm twice. One for getting white columns using the top node, and getting the rows using the left node.